Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

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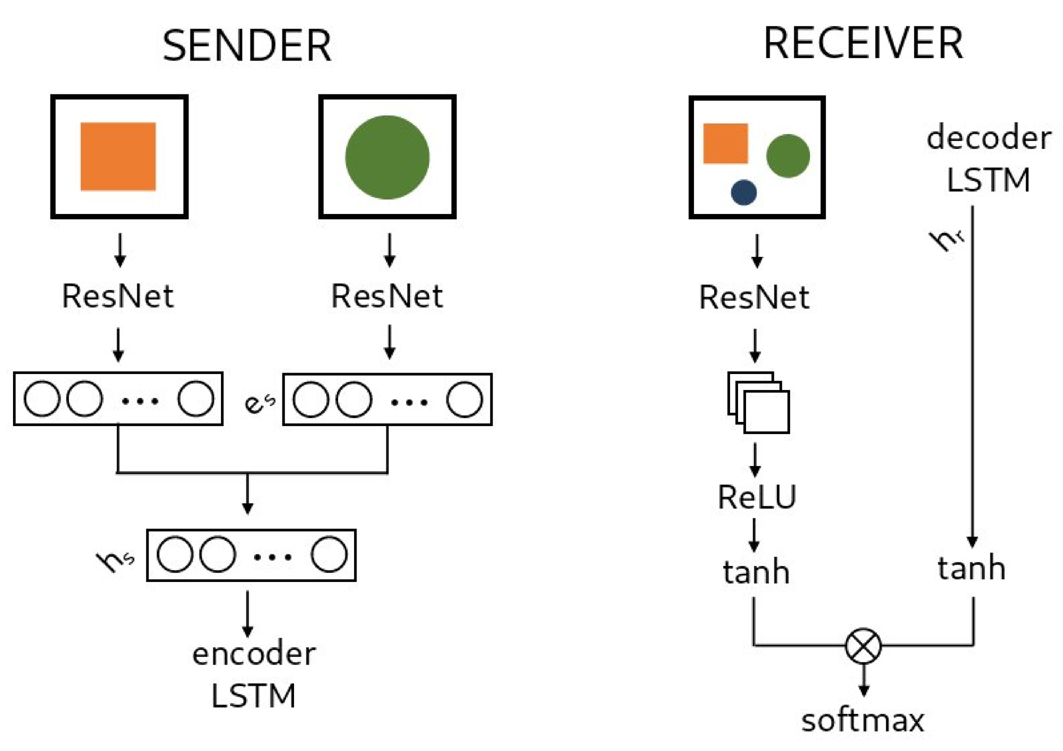
Aims

A picture is worth a thousand words...

- Study **emergence of discrete linguistic symbols** and **languages** from continuous sensory input in **interactive communication**
- Study **constraints** on symbol grounding in interacting neural agents
- Investigate the effects of **structured biases** of **scene complexity**, **discriminating features** present, and **message space** on emergent languages

Setup



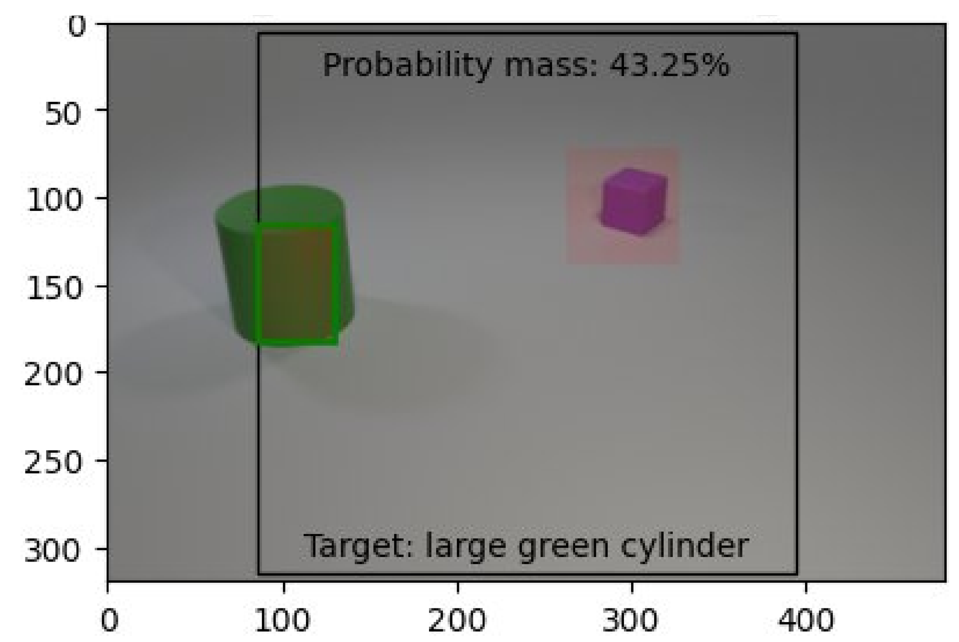
- Receiver **attends to region** in visual scene **based on the sender's message**

Findings

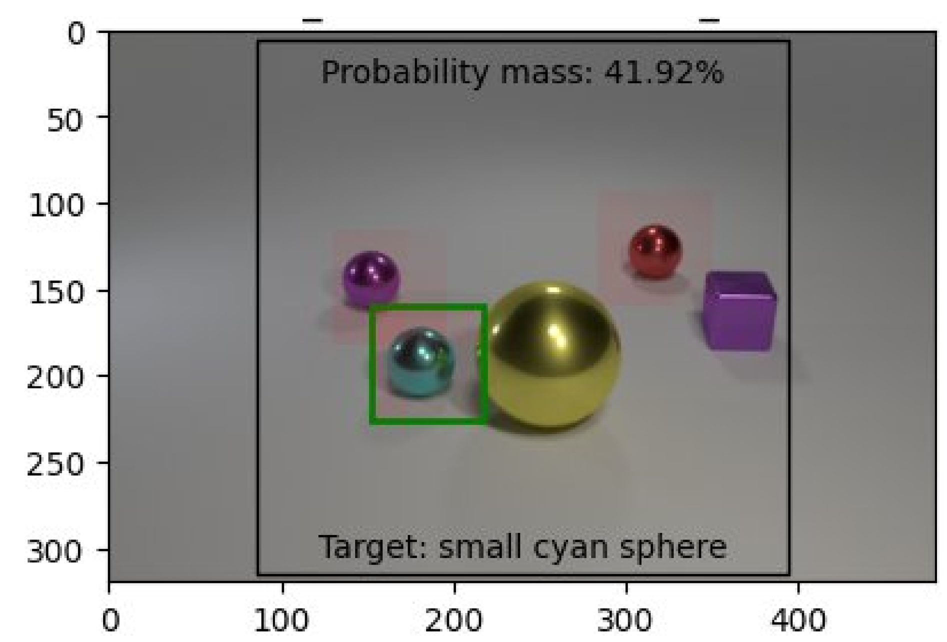
- **Medium vocabularies** ($|V| \in \{10, 16\}$) and message lengths ($n \in \{2, 4\}$) perform best
- Emergent vocabularies **align with the structured contextual biases** introduced
- Scene complexity (**number of distractors, number of features**) strongly impacts communication success
- **Shape** attributes **easiest** to learn, **size** attributes **most challenging**
- **Future:** Compare emergent vocabularies and languages with (different) natural languages

Experiments

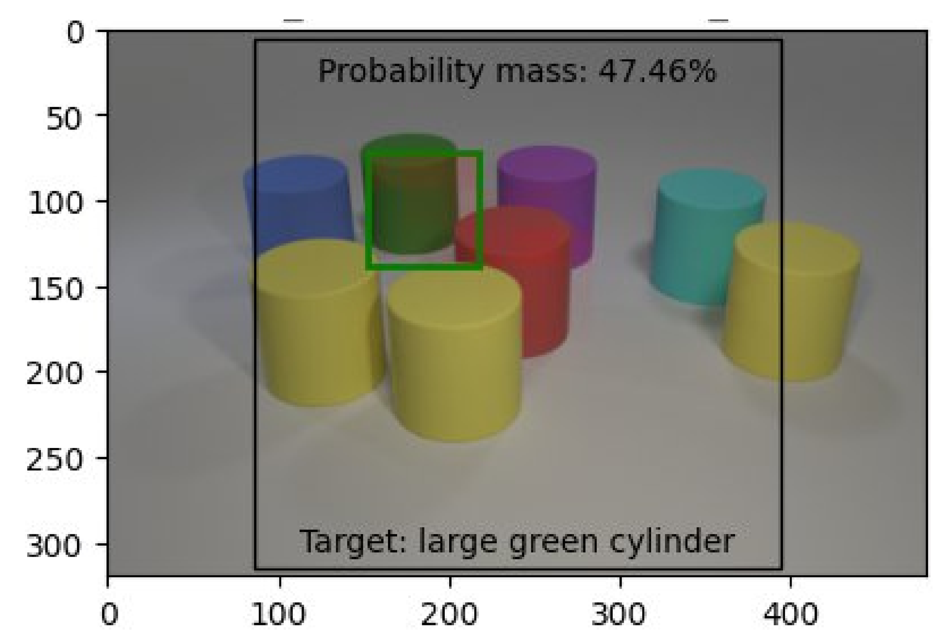
Dale-2



Dale-5



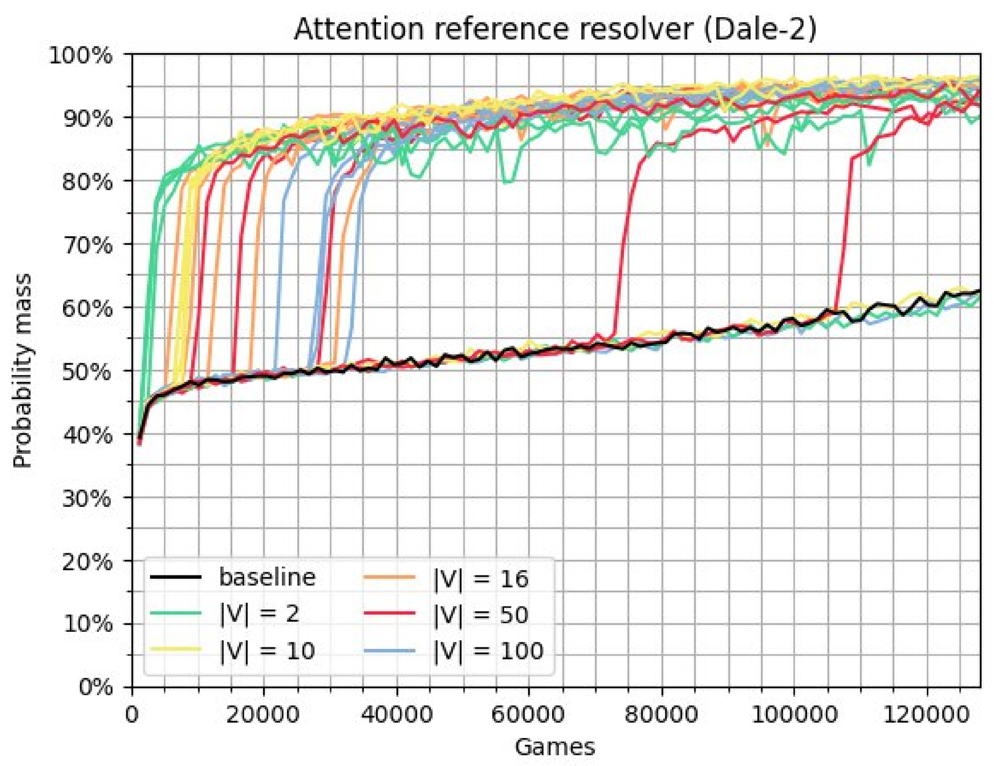
CLEVR Color

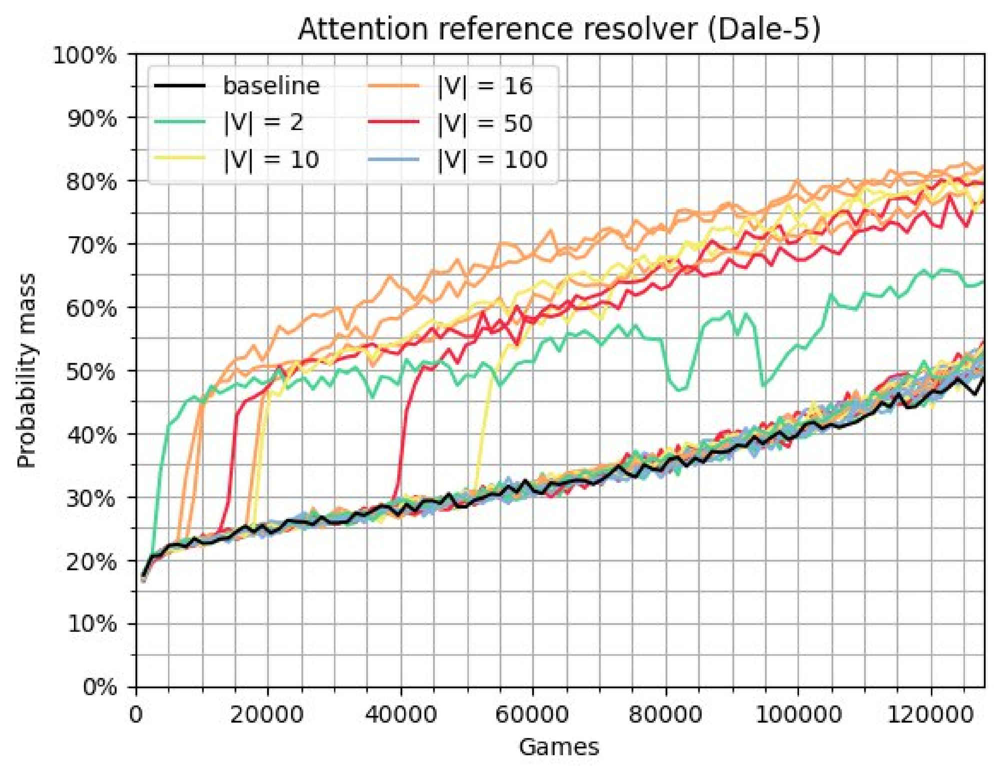


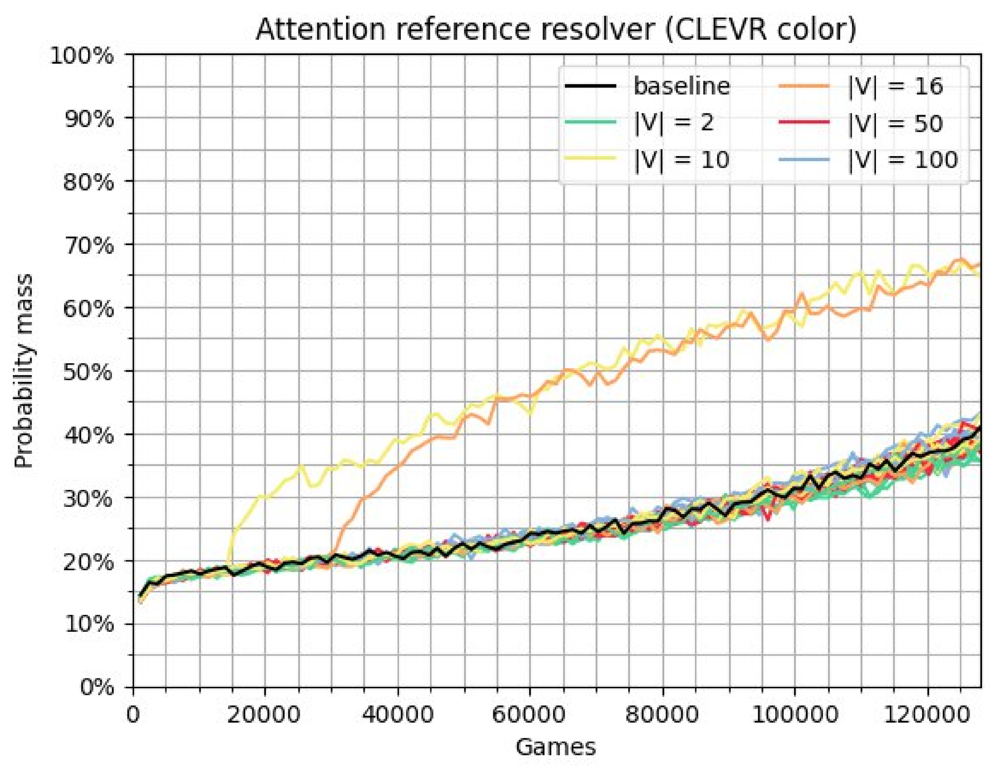
Sender in Supervised Generation (GRE)

Strict string accuracy	99%	69%	93%
F1 score	98,57%	89,53%	95,17%
Shape Prec / Rec	99,75% / 99,74%	98,30% / 98,3%	100% / 100%
Color Prec / Rec	98,09% / 98,53%	92,44% / 92,95%	92,76% / 92,76%
Size Prec / Rec	98,73% / 95,9%	69,43% / 64,13%	N/A
<pad> Prec / Rec	99,64% / 99,77%	82,22% / 84,59%	100% / 100%

Sender and Receiver in Emergent Interaction







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Beyond Pixels and Words

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